LAB 3 REPORT

Component Modelling & Architectural Pattern Selection

Project: Warehouse Inventory & Pallet Location Tracker

Problem Statement: #27 | Smart Cities, Transport & Logistics

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Selected Architecture: Layered Architecture

1. Objective

To evaluate suitable architectural styles for the Warehouse Inventory & Pallet Location Tracker, select an appropriate architecture, and model the system using a UML Component Diagram showing major components, interfaces, dependencies, and interactions.

2. Project Context

The Warehouse Inventory & Pallet Location Tracker is a logistics platform used by warehouse operators and logistics supervisors to manage pallet placement and stock visibility. It supports pallet arrival registration, barcode/RFID scanning, placement capacity validation, pallet retrieval and relocation, pallet location queries, inventory-level monitoring, and low-stock alert generation.

3. Architectural Style Analysis

Architecture	Structure	Advantages	Limitations
Layered Architecture	Presentation, business, data, and external-integration layers.	Clear separation of concerns; easy to understand, test, and maintain as scan/validation rules evolve.	Can introduce communication overhead as calls pass down through layers.
Microservices Architecture	Independent small services (e.g. scan service, inventory service) deployed separately.	Independent deployment, scaling, and fault isolation for high-traffic scan operations.	Higher operational and distributed-system complexity than the project currently needs.
Client-Server Architecture	Handheld scanners/consoles communicate with a centralized warehouse server.	Centralized control and simple deployment across warehouse floor devices.	Possible bottleneck and single point of failure at the central server.

4. Architecture Selection

Layered Architecture is selected because it clearly separates operator/supervisor-facing presentation functions, the business services implementing scanning, validation, retrieval, and monitoring rules, and the data/external-integration responsibilities for persistence and scanner hardware. It provides sufficient modularity for the current project scope without the additional deployment and distributed-system complexity of Microservices.

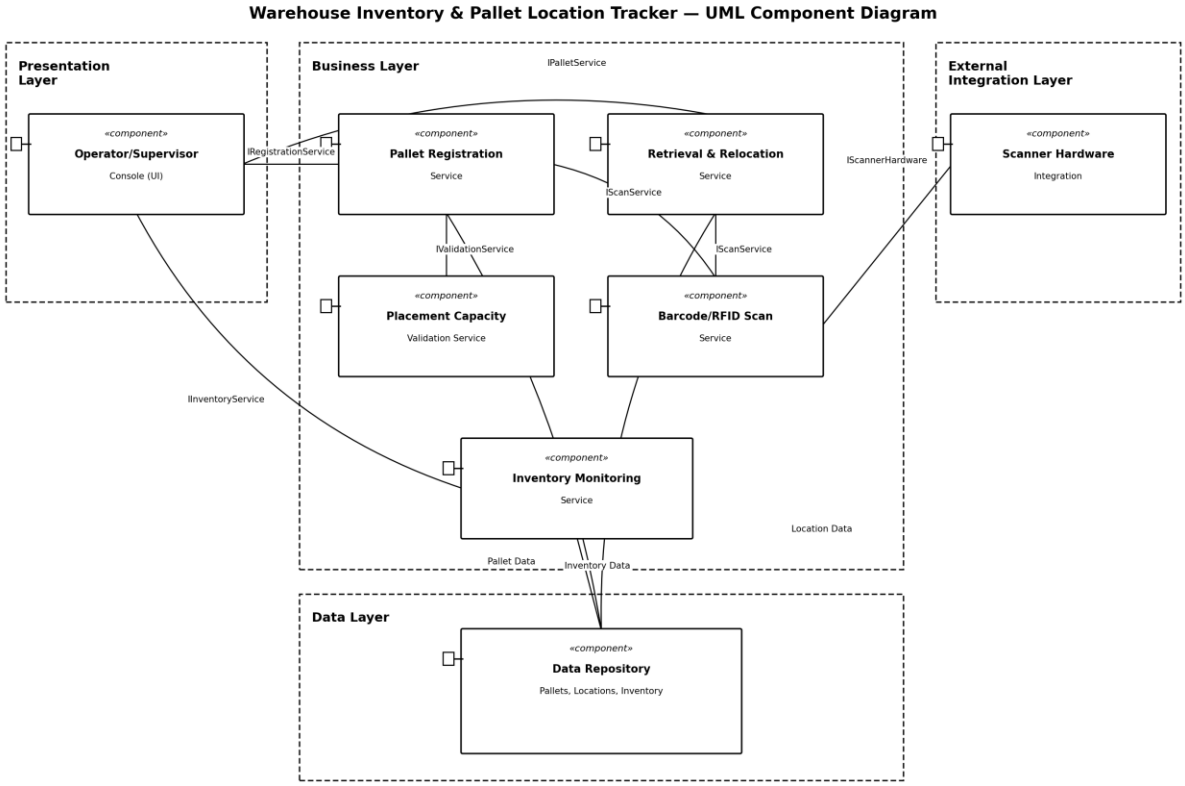
5. Components

Component	Layer	Responsibility
Operator/Supervisor Console (UI)	Presentation	Warehouse operator and logistics supervisor interaction.
Pallet Registration Service	Business	Registers pallet arrivals and initiates placement workflow.

Component	Layer	Responsibility
Retrieval & Relocation Service	Business	Handles pallet retrieval and relocation requests and status.
Placement Capacity Validation Service	Business	Validates shelf/rack load capacity before a pallet is placed.
Barcode/RFID Scan Service	Business	Processes barcode/RFID scans and handles scan failures.
Inventory Monitoring Service	Business	Tracks stock levels and triggers low-stock alerts.
Data Repository	Data	Stores pallet, location, and inventory records.
Scanner Hardware Integration	External	Interfaces with physical barcode/RFID scanner hardware.

6. UML Component Diagram

The final UML Component Diagram is shown below.



7. Interfaces and Interactions

Components	Interface	Purpose
Console ↔ Pallet Registration Service	IRegistrationService	Register Pallet Arrival API
Console ↔ Retrieval & Relocation Service	IPalletService	Retrieve / Relocate Pallet API
Console ↔ Inventory Monitoring Service	IInventoryService	Query Location / Monitor Levels API

Components	Interface	Purpose
Pallet Registration ↔ Validation Service	IValidationService	Placement Capacity Check
Registration / Retrieval ↔ Scan Service	IScanService	Barcode/RFID Scan & Failure Handling
Scan Service ↔ Scanner Hardware Integration	IScannerHardware	Physical Scan Read
Business Services ↔ Data Repository	IDataAccess	Pallet / Location / Inventory Persistence

8. Security Considerations

Sensitive operations are handled by dedicated business services rather than directly by the console UI. Placement capacity validation and successful scan confirmation gate whether a pallet registration, retrieval, or relocation can proceed. Service boundaries also provide clear points for validation and authorization before data reaches the repository.

9. Performance Considerations

Focused components reduce unnecessary processing and keep scanning, validation, and monitoring responsibilities separate. Centralized data access through the Data Repository avoids duplicated persistence logic, while scanner hardware integration is isolated from core registration and retrieval processing so scan failures do not block unrelated operations.

10. Final Justification

Architecture Choice: Layered Architecture was chosen for the Warehouse Inventory & Pallet Location Tracker.

Reason 1: It clearly separates presentation, business logic, and data/integration responsibilities, making the system easier to understand and maintain.

Reason 2: It suits the current project scope and provides modularity without the additional operational complexity of a Microservices architecture.

Security Advantage: Business services consistently control validation, scan confirmation, placement approval, and alert eligibility.

Performance Benefit: The separation of responsibilities keeps operations focused and centralizes data access, reducing unnecessary duplication.

11. Conclusion

The Layered Architecture provides a modular and understandable structure for the Warehouse Inventory & Pallet Location Tracker. The component model supports the major requirements of pallet registration, scanning, placement validation, retrieval and relocation, inventory monitoring, low-stock alerting, and scanner hardware integration.